

Course Code and Title: MAT110 Quantitative Methods
Lesson 1: Week 2
Topic: Data Collection and Presentation
Introduction Data is a systematic record of facts, such as numbers, words, measurements, observations or just descriptions of things from which conclusions may be drawn. They are relevant material, past and present forms as a basis for the research study and analysis. Data collection enables a person or organization to answer relevant questions, evaluate outcomes and make predictions about future probabilities and trends. Accurate data collection is essential to maintaining the integrity of research, making informed business decisions and ensuring quality assurance.
Sources of Data Normally we can gather data from two sources namely primary and secondary. Primary sources provide raw information and give direct access to the subject of research. They are considered to be authoritative because they represent original thinking, reports on discoveries or events, or share new information. Often these sources are created at the time the events occurred but they can also include sources that are created later. They are usually the first formal appearance of original research. Examples include interview transcripts, statistical data, and works of art. Secondary sources provide second-hand information and commentary from other researchers. Examples include journal articles, reviews, and academic books. A secondary source describes, interprets, or synthesizes primary sources. Secondary sources involve analysis, synthesis, interpretation, or evaluation of primary sources. Scholarly journals, although generally considered to be secondary sources, often contain articles on very specific subjects and may be the primary source of information on new developments. A secondary source is anything that describes, interprets, evaluates, or analyzes information from primary sources. When you cite a secondary source, it's usually not to analyze it directly. Instead, you'll probably test its arguments against new evidence or use its ideas to help formulate your own. Common examples include: • Books, articles and documentaries that synthesize information on a topic • Synopses and descriptions of artistic works • Encyclopaedias and textbooks that summarize information and ideas • Reviews and essays that evaluate or interpret something

Primary sources are more credible as evidence, but good research uses both primary and secondary sources. Primary and secondary categories are often not fixed and depend on the study or research you are undertaking. For example, newspaper editorial/opinion pieces can be both primary and secondary. If exploring how an event affected people at a certain time, this type of source would be considered a primary source. If exploring the event, then the opinion piece would be responding to the event and therefore is considered to be a secondary source.

Table 1.0: *Primary and secondary source examples*

Primary source	Secondary source
Novel	Article analyzing the novel
Painting	Exhibition catalog explaining the painting
Letters and diaries written by a historical figure	Biography of the historical figure
Essay by a philosopher	Textbook summarizing the philosopher's ideas
Photographs of a historical event	Documentary about the historical event
Government documents about a new policy	Newspaper article about the new policy
Music recordings	Academic book about the musical style
Results of an opinion poll	Blog post interpreting the results of the poll
Empirical study	Literature review that cites the study

Types of Data

Qualitative data is data that is not given numerically. Data are measures of 'types' and may be represented by a name, symbol, or a number code.

Examples: favorite color, place of birth, favorite food, type of car, gender

Quantitative data is numerical. Data are measures of values or counts and are expressed as numbers.

There are two types of quantitative data:

Discrete data can only take specific numeric values

Example: shoe size, number of brothers, number of cars in a car park

Continuous data can take any numerical value; data have infinite possibilities

Examples: height, mass, length

Quantitative and qualitative data provide different outcomes, and are often used together to get a full picture of a population. For example, if data are collected on annual income (quantitative), occupation data (qualitative) could also be gathered to get more detail on the average annual income for each type of occupation.

Data Collection Methods

Although data can be valuable, too much information is unmanageable, and the wrong data is useless. The right data collection method can mean the difference between useful insights and time-wasting misdirection. Organizations have several tools for primary data collection. The methods range from traditional and simple, such as a face-to-face interview, to more sophisticated ways to collect and analyze data.

1. Interviews

Interviews are used to collect data from a small group of subjects on a broad range of topics. You can use structured or unstructured interviews. Structured interviews are comparable to a questionnaire, with the same questions in the same order for each subject and with multiple choice answers. For unstructured interviews questions can differ per subject and can depend on answers given on previous questions, there is no fixed set of possible answers.

Almost anyone can come up with a list of questions, but the key to efficient interviews is knowing what to ask. Efficiency in interviewing is crucial because, of all the primary data collection methods, in-person interviewing can be the most expensive. There are ways to limit the cost of interviews, such as conducting them over the phone or through a web chat interface. But sometimes an in-person interview can be worth the cost, as the interviewer can tailor follow-up questions based on responses in a real-time exchange.

Interviews also allow for open-ended questions. Compared to other primary data collection methods, such as surveys, interviews are more customizable and responsive.

2. Observation

Observation involves collecting information without asking questions. This method is more subjective, as it requires the researcher, or observer, to add their judgment to the data. But in some circumstances, the risk of bias is minimal. For example, if a study involves the number of people in a restaurant at a given time, unless the observer counts incorrectly, the data should be reasonably reliable. Variables that require the observer to make distinctions, such as how many millennials visit a restaurant in a given period, can introduce potential problems.

In general, observation can determine the dynamics of a situation, which generally cannot be measured through other data collection techniques. Observation also can be combined with additional information, such as video.

3. Documents and records

Sometimes you can collect a considerable amount of data without asking anyone anything. Document and records-based research uses existing data for a study. Attendance records, meeting minutes, and financial records are just a few examples of this type of research. Using documents and records can be efficient and inexpensive because you're predominantly using research that has already been completed. However, since the researcher has less control over the results, documents and records can be an incomplete data source.

4. Focus groups

A combination of interviewing, surveying, and observing, a focus group is a data collection method that involves several individuals who have something in common. The purpose of a focus group is to add a collective element to individual data collection. A focus group study can ask participants to watch a presentation, for example, then discuss the content before answering survey or interview-style questions.

Focus groups often use open-ended questions such as, "How did you feel about the presentation?" or "What did you like best about the product?" The focus group moderator can ask the group to think back to the shared experience, rather than forward to the future.

5. Questionnaires and surveys

Questionnaires and surveys can be used to ask questions that have closed-ended answers. Data gathered from questionnaires and surveys can be analyzed in many different ways. You can assign numerical values to the data to speed up the analysis. This can be useful if you're collecting a large amount of data from a large population.

To be meaningful, surveys and questionnaires need to be carefully planned. Unlike an interview, where a researcher can react to the direction of a respondent's answers, a poorly designed questionnaire will lead the study nowhere quickly. While surveys are often less expensive than interviews, they won't be valuable if they aren't handled correctly.

Surveys can be conducted as interviews, but in most cases, it makes sense to conduct surveys using forms. Online forms are a modern and effective way to conduct surveys. Unlike written surveys, which are static, the questions presented in online forms can change according to how someone responds.

6. Experimentation

A designed experiment is a controlled study whose purpose is to control as many factors as possible to isolate the effects of a particular factor. Designed experiments must be carefully set up to achieve their purposes. The variables in a designed experiment that are controlled are called the explanatory variables or are sometimes called the factors. Factors have values that can be changed by the researcher and are considered as possible causes. Examples of factors are:

- The dosage of a drug in a medical experiment
- The type of teaching method in an education experiment
- One drug by itself compared with that drug used in conjunction with another

The designed experiment analyzes the effects of the factors on the response variable. Response variables are not part of a controlled environment and have values that are measured by the researcher. Examples of response variables are:

- The blood pressures of the patients
- The test scores for a class
- The sizes of a cancerous tumor for patients

A treatment is the specific combination of the values of the factors. Examples of treatments are:

- Giving one medication to one group of patients and a different medication to another
- Using one type of fertilizer on a set of plots of corn and a different type of fertilizer on a different set of plots
- Playing country music to one group of mice and rap music to another

A treatment is applied to "experimental units" (people, plants, materials, or other objects). When experimental units are people, we refer to them as subjects. Subjects in an experiment correspond to

individuals in a survey.

Data Presentation

Once data has been collected, it has to be classified and organized in such a way that it becomes easily readable and interpretable that is, converted to information. Before the calculation of descriptive statistics, it is sometimes a good idea to present data as tables, charts, diagrams or graphs. Most people find pictures much more helpful than numbers in the sense that, in their opinion, they present data more meaningfully.

1. Tabular Forms

This type of information occurs as individual observations, usually as a table or array of disorderly values. These observations are to be firstly arranged in some order (ascending or descending if they are numerical) or simply grouped together in the form of a frequency table before proper presentation on diagrams is possible.

1.1 Arrays

An array is a matrix of rows and columns of numbers which have been arranged in some order (preferably ascending). It is probably the most primitive way of tabulating information but can be very useful if it is small in size. Some important statistics can immediately be located by mere inspection.

Without any calculations, one can easily find the:

1. Minimum observation
2. Maximum observation
3. Number of observations, n
4. Mode
5. Median, if n is odd

2	7	8	11	15
16	18	19	19	19
23	23	24	26	27
29	33	40	44	47
49	51	54	63	68

Minimum = 2

Maximum = 68

Number of observations = 25

Mode = 19

Median = 24

1.2 Simple Tables

A table is slightly more complex than an array since it needs a heading and the names of the variables involved. We can also use symbols to represent the variables at times, provided they are sufficiently explicit for the reader. Optionally, the table may also include totals or percentages (relative figures).

Distribution of Ages of Students	
Age of student	Frequency
19	14
20	23
21	134
22	149
23	71
24	9
Total	400

1.3 Compound Tables

A compound table is just an extension of a simple table in which there are more than one variable distributed among its attributes (sub- variable). An attribute is just a quality, property or component of a variable according to which it can be differentiated with respect to other variables.

		DEGREE		
		BSBA	BSIT	BSEE
RE	Passed	37	25	33
SU				

Table 1.2: Example

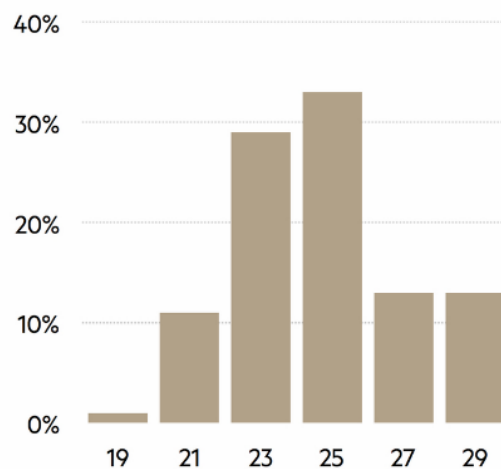
	Failed	13	8	15
LT	Incomplete	5	3	10

2. Graphs

Graphs display data and should be chosen according to the type of data being presented. Commonly used graphs include bar- and pie-charts, histograms, box-and-whisker plots and scatter diagrams. Graphs can be symmetrical, positively skewed or negatively skewed. Be aware that graphs may be presented in misleading ways.

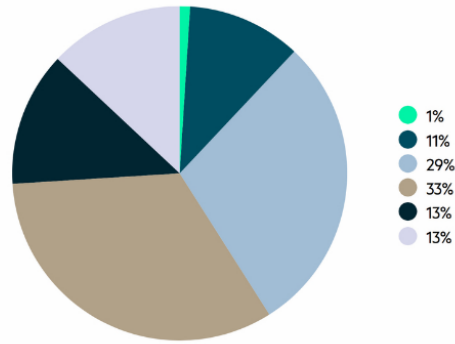
2.1 Bar-chart

A bar-chart can be used for nominal data, ordered data or, if there are only a few categories, whole numbers. The bars should be spaced apart, with each height being proportional to the category's percentage out of the total number of observations.



2.2 Pie-chart

Pie-charts operate in a similar manner but use sectors within a circle; the angle of each sector is proportional to the category's percentage out of the total number of observations.



2.3 Histogram

With quantitative data, histograms should be used rather than bar-charts. Bars should not have gaps between them as each represents an interval for the variable under consideration.

The histogram is constructed using the number of observations in each interval. It is good practice to use intervals of equal width, as the height of each bar is then proportional to the category's percentage out of the total number of observations. The number of intervals to be used should be considered carefully. Too few intervals may prevent important features of the distribution from being identified, whereas too many leads to a distribution with a jagged rather than a smooth shape.

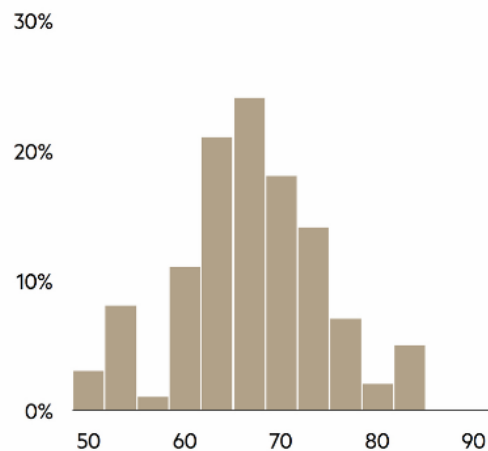
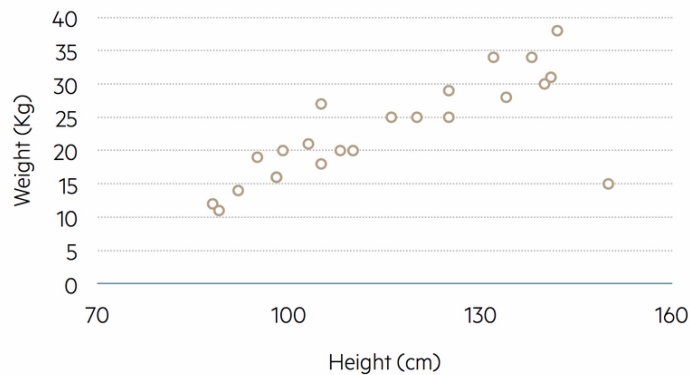


Figure 1.1

2.4 Scatter diagram

The scatter diagram graphs pairs of numerical data, with one variable on each axis, to look for a relationship between them. If the variables are correlated, the points will fall along a line or curve. The

better the correlation, the tighter the points will hug the line. This cause analysis tool is considered one of the seven basic quality tools. A scatter diagram is used when; you have paired numerical data, when your dependent variable may have multiple values for each value of your independent variable and when trying to determine whether the two variables are related.



2.5 Line plots

Trends over time can be illustrated as line plots. For instance, the number of cases of measles for consecutive years can be shown as points with the numbers represented by the vertical axis, the base set at zero, and the years indicated along the horizontal axis. Adjacent points are joined together by straight lines to highlight trends.

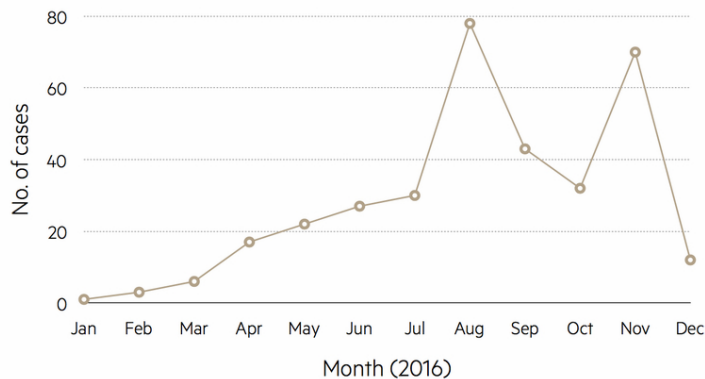


Figure 1.3: Ex

Summary

Data are facts and statistics collected together for reference or analysis. There are two sources of

data, the primary source and the secondary source of data. Primary sources provide raw information and give direct access to the subject of research, while secondary sources provide second-hand information and commentaries from other researchers. There are two types of data, the quantitative and the qualitative datum. Quantitative data are measures of values or counts and are expressed as numbers, while qualitative data are measures of types and are expressed as names, symbols, or number codes.

There are different methods of collecting data. They can be collected using interviews, observations, documents and records, focus groups, questionnaires and surveys, and experimentation. Once data has been collected, it has to be classified and organized in a systematic way to make them readable and interpretable. Presentation of data can be tables or graphs. For tabular forms, there are arrays, simple tables, and compound tables, and for graphs, there are bar charts, pie charts, histograms, scatter diagrams, and line plots.